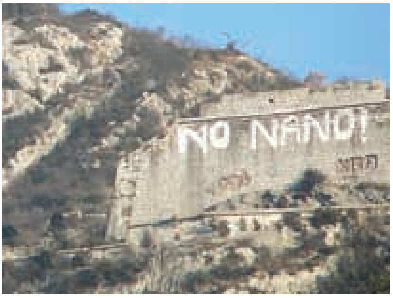
ficult to generalise about health risks associated with exposure to nanomaterials – each new nanomaterial must be assessed individually and all material properties must be taken into account.

Environment

As nanotechnology is an emerging field, there is great debate regarding to what extent industrial and commercial use of nanomaterials will affect organisms and ecosystems. Nanopollution caused by the waste generated by nanodevices or during their manufacturing process, might easily penetrate animal and plant cells causing unknown effects.



NO NANO! Grafitti on the Bastille fortress in Grenoble, France

Living organisms do not have a mechanism to deal with artificially engineered nanowaste. They do not appear in nature, and so no evolutionary changes have adapted our immune system to this new threat. This makes nanopollution all the more dangerous.

Society

The "societal implications of nanotechnology" is a term used to include all the potential benefits and challenges that the introduction of novel nanotechnological devices and materials may hold for society and human interaction.

A technology with the potential to revolutionise manufacturing, health care, energy supply, communications and probably defense, will also transform labour and the workplace, the medical system, the transportation and power infrastructures and the military. None of these will be changed without significant social disruption. In fact, many of the most enthusiastic proponents of nanotechnology, see this nascent science (Nanoscience) as a mechanism to changing human nature itself – going beyond curing disease and enhancing human characteristics.

Ethics

As lifesaving tools, nanotechnology is unsurpassed in its promise of an absolute revolution for medical treatment of previously incurable or untreatable conditions. Conversely, when this technology is used to manufacture miniature weapons or explosives the infinite possibilities of far-reaching reverberations is a very